

Introduction

Diagnosis &
Screening

Testing &
counseling

Systemic Tx

Cases

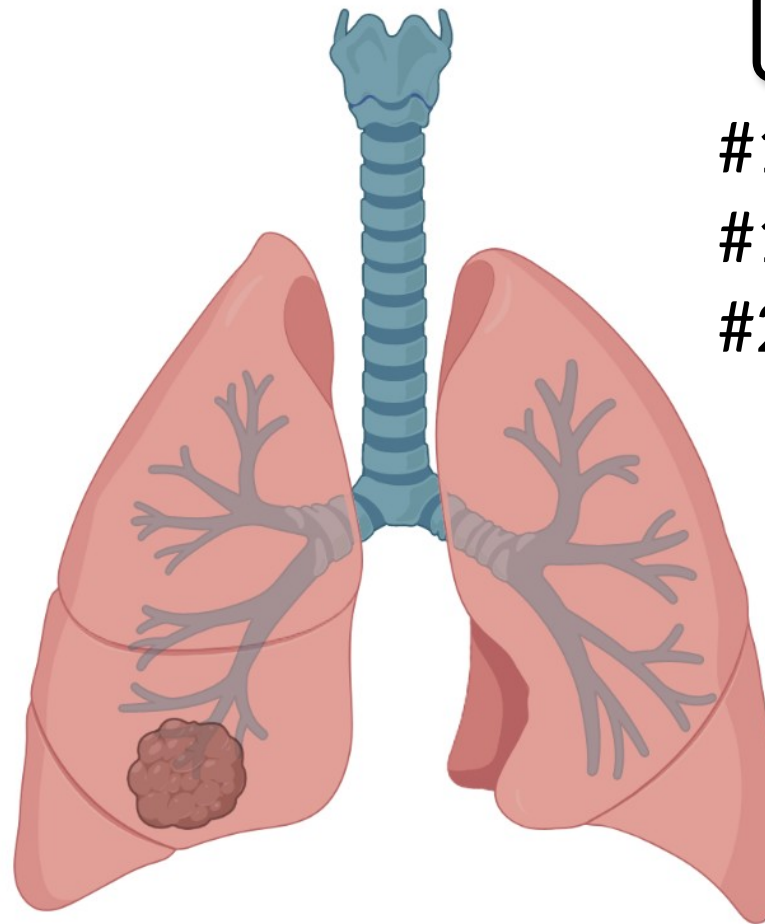
Lung Cancer: Diagnosis and First Steps for the Internist

Importance and Impact?

#1 in cancer deaths worldwide

#1 cause of cancer in men

#2 cause of cancer in women



Introduction

Diagnosis & Screening

Asymptomatic

Symptomatic

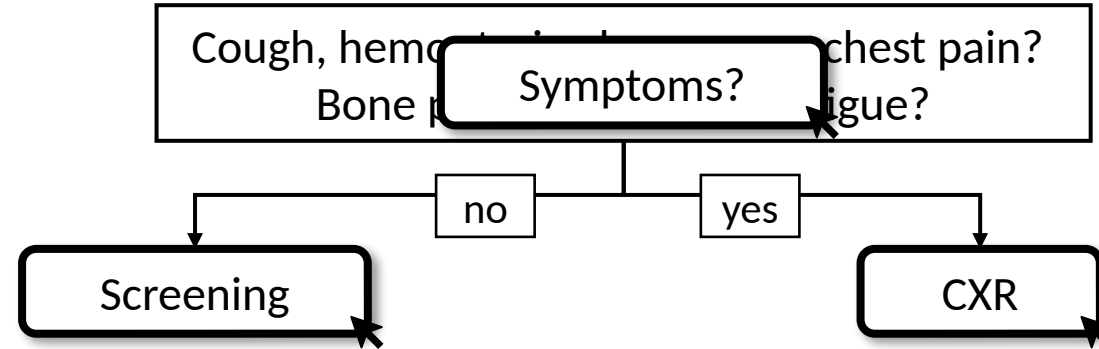
Biopsy

Testing & counseling

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Cases

Diagnosis & Screening



Introduction

Diagnosis & Screening

Asymptomatic

Symptomatic

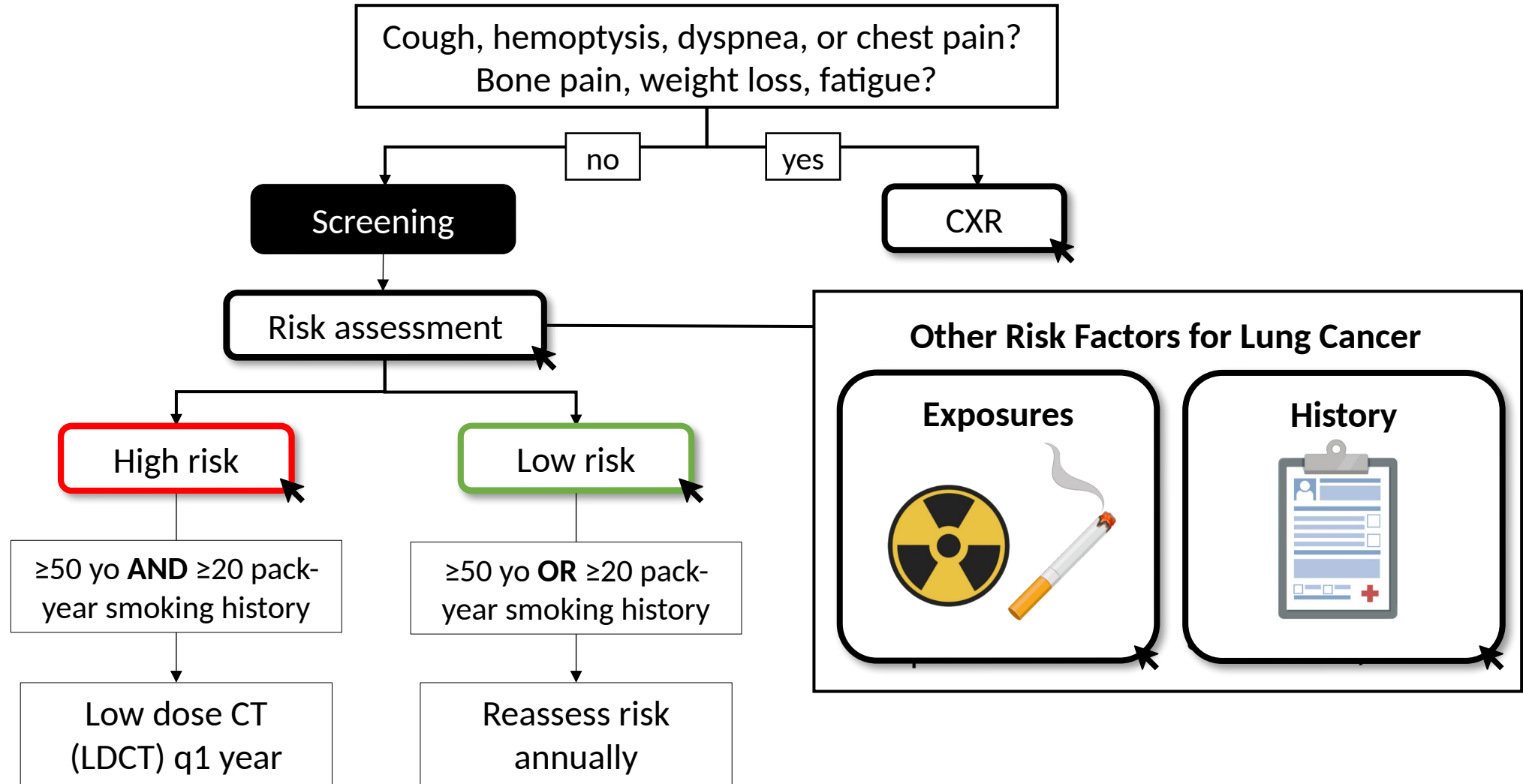
Biopsy

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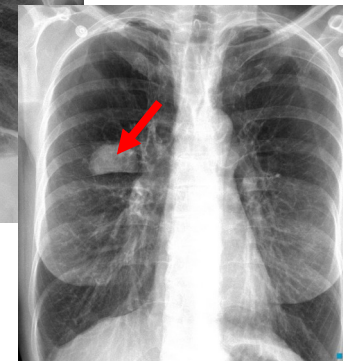
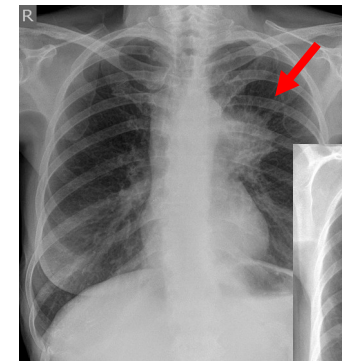
Cough, hemoptysis, dyspnea, or chest pain?
Bone pain, weight loss, fatigue?

no

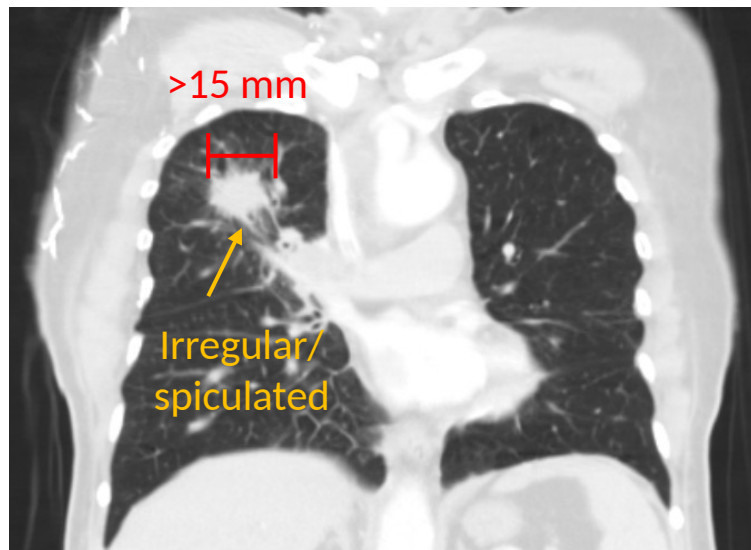
yes

Screening

CXR



CT Findings Concerning For Malignancy



- Thick-walled cavitation
- Growth
- Solid component in ground glass
- Extrapulmonary lesions

Concerning

Not concerning

Risk assessment

CT Chest +/-
Abd/Pelvis

High risk
OR
Persistent sx

Low risk & sx
improve

No further
work-up

Biopsy

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Where and How to Biopsy?

Click on each biopsy sites to determine which specialist would perform the biopsy.

Peripheral lung

(IR/Surgery)

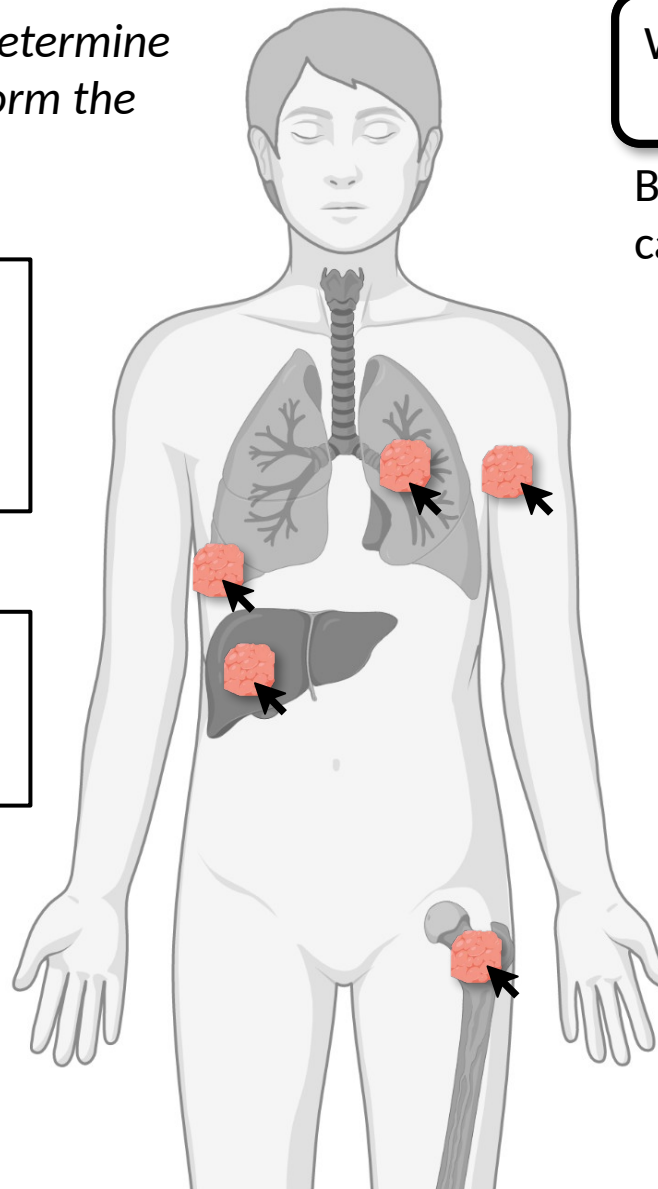
May be amenable to CT guided biopsy

Hepatic lesions (IR)

May be amenable to CT/US guided biopsy

Bone lesions (Ortho/IR)

Can be biopsied by orthopedic surgery or IR



Which lesion would you biopsy to make a diagnosis?

Biopsy most advanced lesion as it can give you diagnosis and staging.

Endobronchial (Pulmonology)

Masses near airways may be biopsied by bronchoscopy

Skin/ soft tissue / lymph nodes (General surgery/ IR)

Excisional lymph node biopsy is typically done by general surgery. CT/US guided punch/core biopsy is done by IR.

Lung Cancer: Diagnosis and First Steps for the Internist

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SCLC

NSCLC

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Biopsy-proven diagnosis

Staging
(extent of cancer)

- CT CAP + IV contrast OR PET-CT
- MRI brain with and without contrast
- Bronchoscopy +/- mediastinoscopy*

*varies with initial staging; typically when considering surgical resection

Assessment of surgical candidacy

- PFTs*

Additional labs?

Assess metastatic involvement

- CBC
- BMP + Ca
- LFTs

Determine targeted therapy
(NSCLC only)

- PD-L1 percentage
- Molecular testing

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Overview



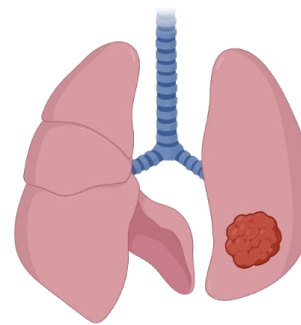
10-15% of all lung cancers

- Smoking is major RF
- Rapidly progressive
- Brain metastases
- Associated with paraneoplastic syndromes

Small Cell Lung Cancer

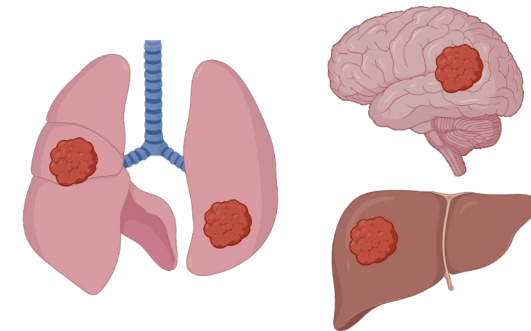
Staging

Limited Stage



Confined to ipsilateral chest that can be treated with single radiation field

Extensive Stage



Spread beyond ipsilateral chest
Distant metastases

Treatment



OR



Treatment



5-year survival rate

Median survival

10-13%

15-20 mos

Prognosis

1-2%

8-13 mos

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Non-Small Cell Lung Cancer

Overview



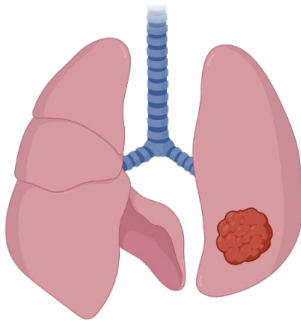
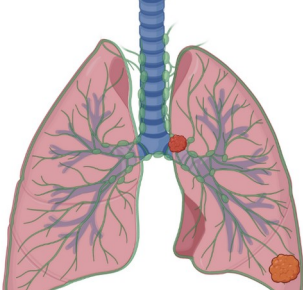
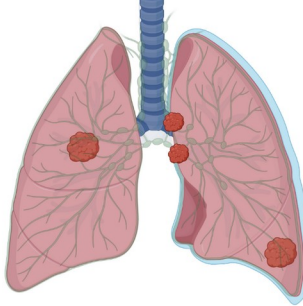
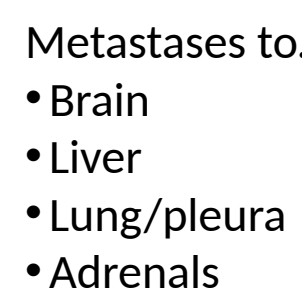
80-85% of all lung cancers

- Histological subtypes:
 - Adenocarcinoma
 - Squamous
 - Other
- Treatment dependent on histology and molecular testing

Treatment

5-year survival rate
Median survival

Staging

I	II	III	IV
			
 OR 	Treatment  OR 		
Prognosis 56-73-90% 66 mos		12-41% 11-41.9 mos	0-10% 6-11.5 mos

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Characteristics

Staging

Treatment

Prognosis

SCLC vs NSCLC

SCLC



10-15% of all lung cancers

- Smoking associated
- Brain mets
- Paraneoplastic syndromes

Limited vs Extensive Stage

Depends on stage



Immuno-/ chemotherapy

Poor

NSCLC



80-85% of all lung cancers

- Different histological subtypes
- Molecular characteristics important

Stage I - IV

Depends on stage and molecular markers



Immuno-/ chemo-/ targeted therapy

Good (stage I) to Poor (stage IV)

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- Cases

Systemic Therapies

Chemotherapy

Indication

Carboplatin/ Cisplatin

Examples

Etoposide
Docetaxel/ Paclitaxel
Lurbinectedin

Nausea/ vomiting/ diarrhea

Toxicities

Nephropathy (AKI)

Cytopenias

Hair loss

Immunotherapy

Indication

Pembrolizumab

Examples

Durvalumab

The “-itises”

pneumonitis, thyroiditis, colitis, myocarditis

Targeted Therapy

Indication

EGFR { Osimertinib
Erlotinib

Examples

ALK { Crizotinib

ROS1 { Crizotinib
Entrectinib
Lorlatinib

Anemia

Toxicities

Kasri

GI symptoms

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Case 1

47-year-old with history of hyperlipidemia, HTN, CAD who presents to establish care. They have no complaints today. Social history is notable for a 20 pack-year smoking history. Strong family history of lung cancer in his mother, father, and paternal grandparent – all of whom were smokers.

What are their risk factors for lung cancer?

Smoking history/pack years, family history of lung cancer

Would you screen for lung cancer?

No. While they are at higher risk based on family history and smoking history, screening for asymptomatic patients with high risk of lung cancer is recommended starting at age 50.

How would you counsel this patient?

Emphasize smoking cessation, low dose CT in 3 years at 50

Case 2

The patient represents 5 years later (at age 52) with 3 months of cough and 2 weeks of hemoptysis. They additionally report fatigue, 20 lb weight loss in 6 months, and persistent right upper quadrant pain. They continue to smoke 1 pack per day.

What is your next step in evaluation?

For symptomatic patients, a CXR can be the first step in evaluation. Additional laboratory evaluation (e.g., CBC, BMP, LFTs) to evaluate for other causes of symptoms.

...CXR shows 2.5 cm ill defined nodule.

What is your next step in evaluation?

CT chest + abdomen/pelvis with IV contrast for staging

...CT chest, abdomen, pelvis shows 2.5 cm spiculated nodule with multiple hypodensities in the liver.

What is your next step in evaluation?

Consult IR for liver biopsy. This will aid in diagnosis and staging.

Lung Cancer Screening - 2023-2025 Update

What's changed since this talk (Dec 2023)

Screening — ACS 2023 guideline (Wolf, CA Cancer J Clin 2024): annual LDCT for adults 50-80 yo with ≥ 20 pack-year history who currently OR formerly smoke. Key change: the "quit within 15 years" cutoff was REMOVED -- years since quitting no longer limits eligibility to start or stop. (USPSTF 2021 -- still the basis for most coverage -- keeps the ≤ 15 -yr-quit rule: age 50-80, ≥ 20 pack-yr, current or quit ≤ 15 yr.)

Resectable NSCLC now gets immunotherapy around surgery -- neoadjuvant nivolumab + chemo (CheckMate 816; FDA Mar 2022) and perioperative pembrolizumab (neoadj + adjuvant; KEYNOTE-671; FDA Oct 2023). Adjuvant atezolizumab (IMpower010) and perioperative durvalumab (AEGEAN) also approved.

EGFR-mutated resected NSCLC (stage IB-IIIa): adjuvant osimertinib is standard of care (ADAURA; FDA Dec 2020; overall-survival benefit reported 2023). Reinforces the deck's point -- molecular testing changes both treatment AND prognosis, now even in early-stage disease.

Standing teaching unchanged: the screening risk-assessment framework, symptomatic CXR->CT pathway, biopsy-the-most-advanced-lesion rule, full staging workup, and SCLC-vs-NSCLC contrasts all remain current.

Lung Cancer Screening

Take-Home Points

Screen the right people: annual low-dose CT for 50-80 yo with ≥ 20 pack-years (USPSTF: current or quit ≤ 15 yr; ACS 2023 drops the quit-time limit). Asymptomatic patients get risk assessment; symptomatic patients start with a CXR.

Tissue is the issue -- and biopsy the MOST advanced lesion when safe, because one sample gives you diagnosis AND staging (a distant met = stage IV).

Stage fully before treating: CT chest/abd/pelvis or PET-CT, MRI brain, node sampling, PFTs if a surgical candidate, plus PD-L1 and molecular testing.

Molecular markers (EGFR, ALK, ROS1) and PD-L1 steer targeted therapy and immunotherapy -- ~50% of never-smokers harbor a targetable mutation. Early stage can be cured; survival falls sharply with each stage, which is why screening matters.

Lung Cancer Screening

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Adapted from TeachIM.org -- "Lung Cancer Screening" (Hematology). Author: Yunan Nie, MD; Section Editor: Chris Geiger, MD; Executive Editors: Yilin Zhang, MD and Brandon Fainstad, MD. Published December 2023.

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Optimization (June 2026): Guideline currency verified. No factual errors found in the original deck -- this is an append-only edit; all 11 original animated/click-to-reveal slides and BioRender graphics are preserved unchanged. Three slides appended: 2023-25 update, take-home points, and this attribution. Newer context added (all source-cited): ACS 2023 screening expansion; perioperative/neoadjuvant immunotherapy for resectable NSCLC (CheckMate 816, KEYNOTE-671); adjuvant osimertinib for EGFR-mutated resected NSCLC (ADAURA).

Companion files: Lung_Cancer_Screening_Lesson_Package.html (launcher), _OnePager.pdf, _Anki_Cards.txt. Original teaching guide = the deck itself (printable version).